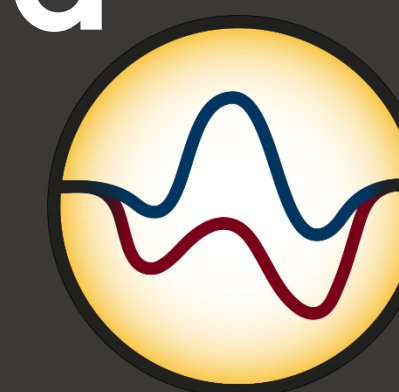


Is photon enhancement of catalytic rates limited by kinetics of photochemical processes?



Center for
Programmable
Energy Catalysis

Silvia Marino, Arli Andrews, Ali Mohammed, Omar Abdelrahman,
Michael Gordon and Phillip Christopher*

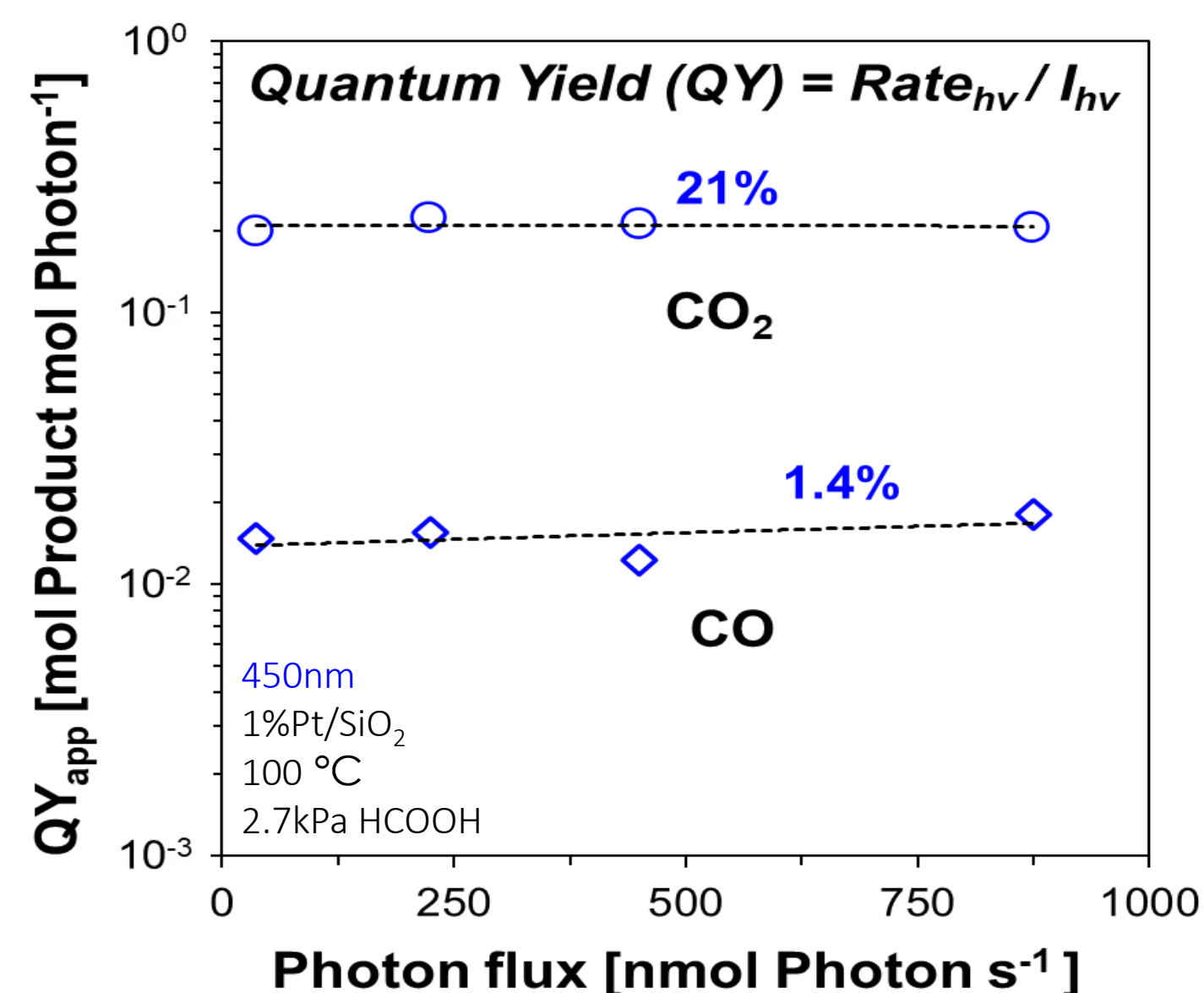
Why is QY_{CO_2} higher than QY_{CO} ?

Formic acid
decomposition

HCOOH



**CO_2 formation
rate ~ 20x faster
than CO rate**

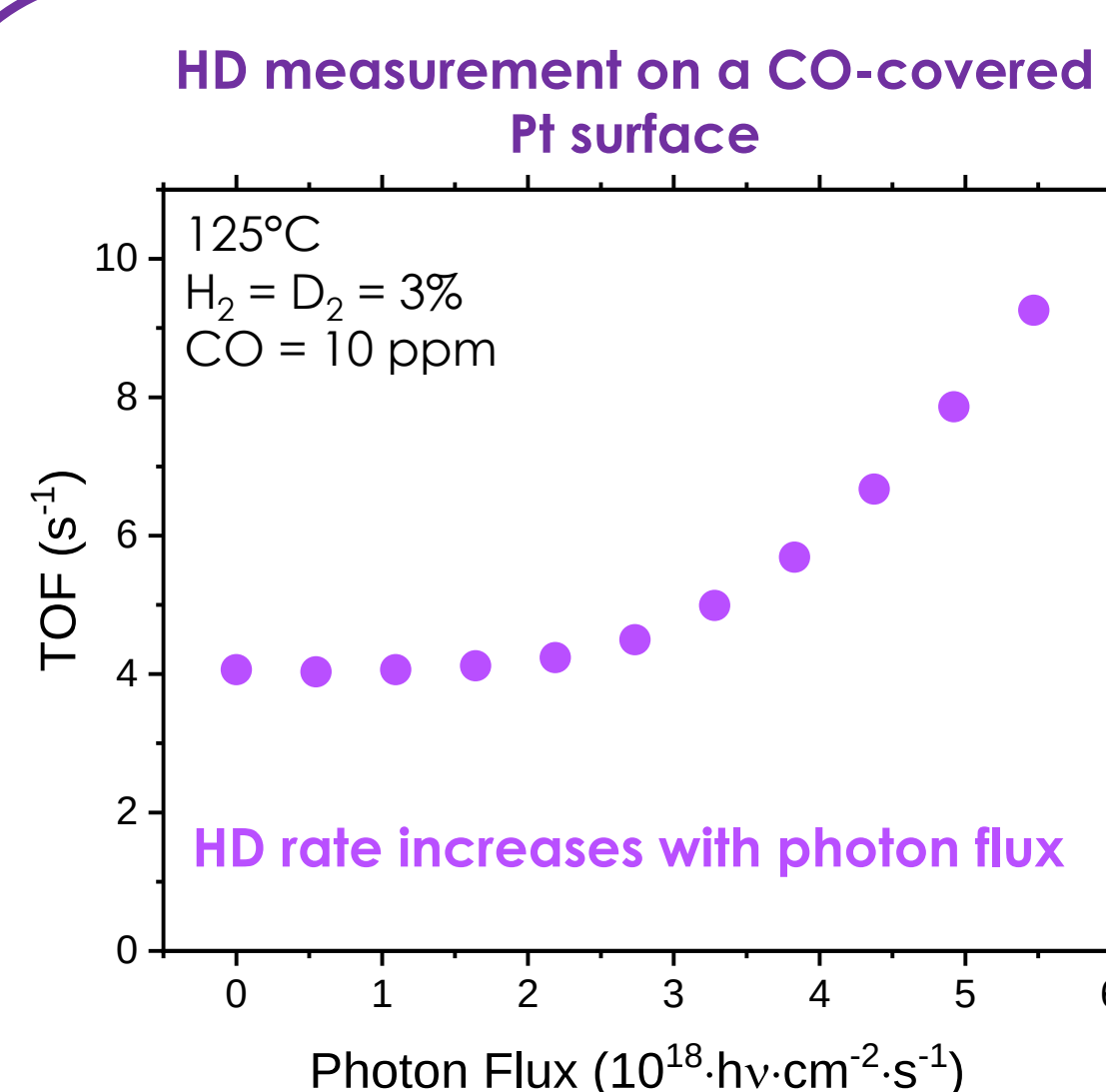


Gain estimation

$$\text{Gain} = \frac{\text{rate}_{HD}}{\text{rate}_{CO_photo_pair_sites}}$$

Experiments

Modeling

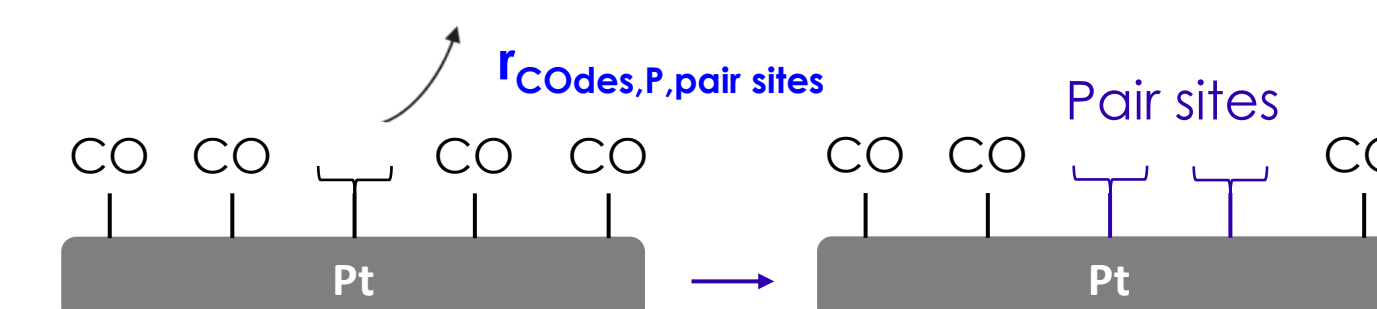


Estimation of photo-desorption rate to form pair sites

$$\frac{d\theta_{CO*}}{dt} = k_{ads}\theta_*\left(\frac{\theta_{CO*}}{\theta_*}\right) - (k_{des,T} + k_{des,P})\theta_*\left(\frac{\theta_{CO*}}{\theta_*}\right)$$

$$k_{des,P} = J k_{AQE} \left(-\frac{E_{photo}}{RT} \right)$$

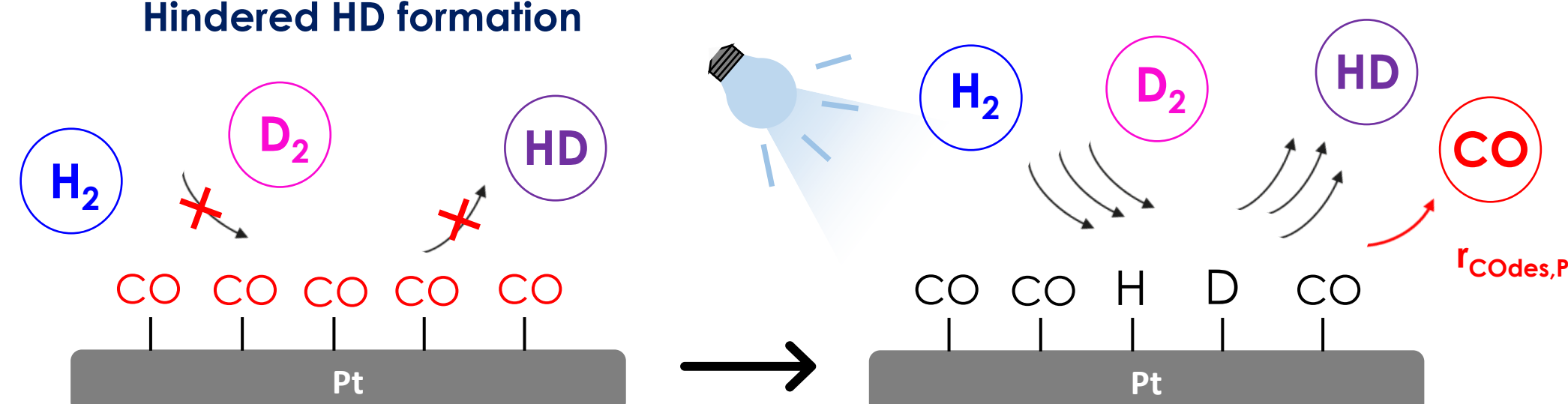
$$\text{rate}_{CO_photo_pair_sites} = k_{des,P}\theta_{CO*}$$



H_2 - D_2 as a model system

$$\text{Gain} = \frac{\text{observed rate increase} - \text{not directly affected by light}}{CO \text{ photodesorption rate}}$$

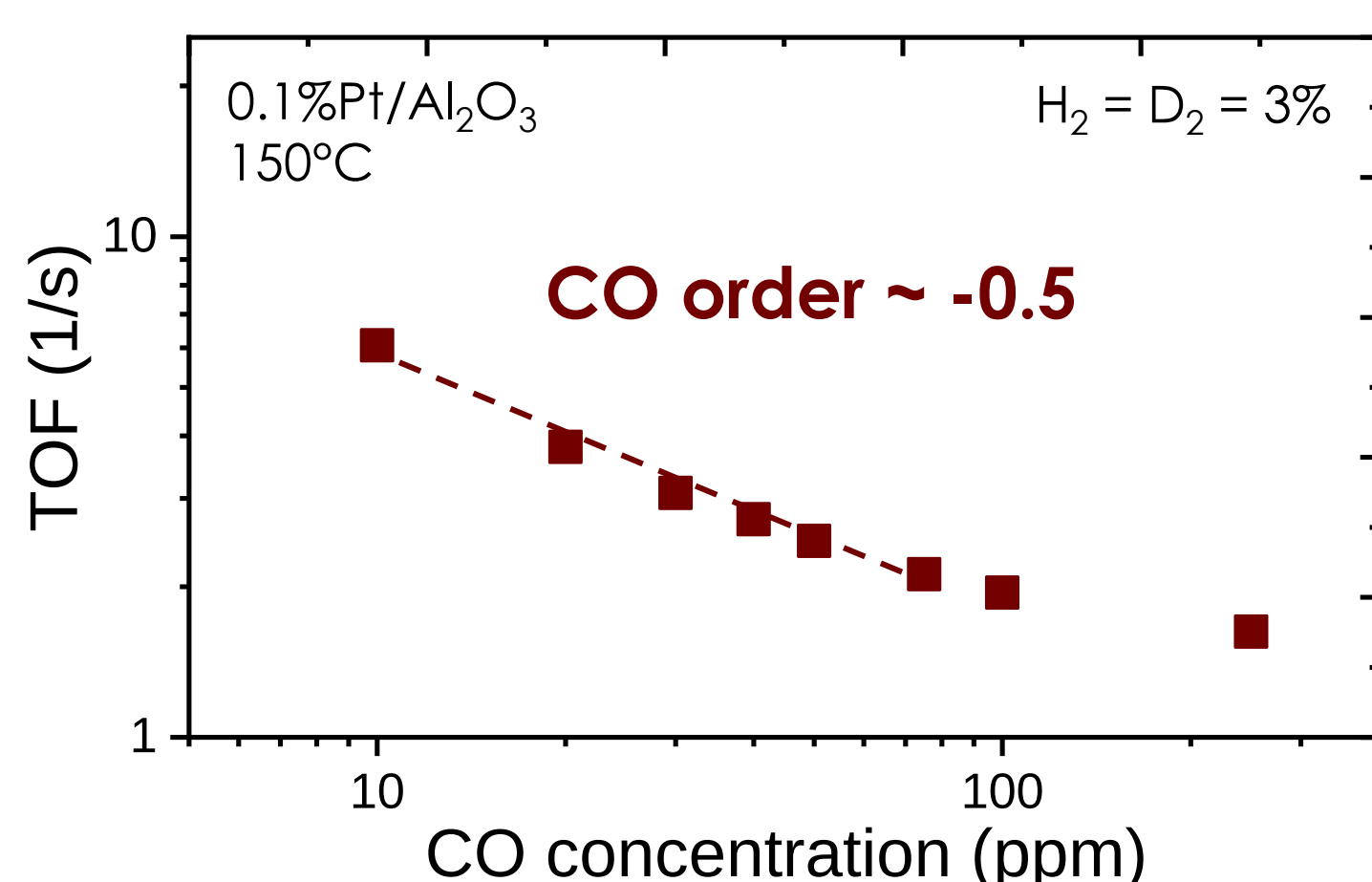
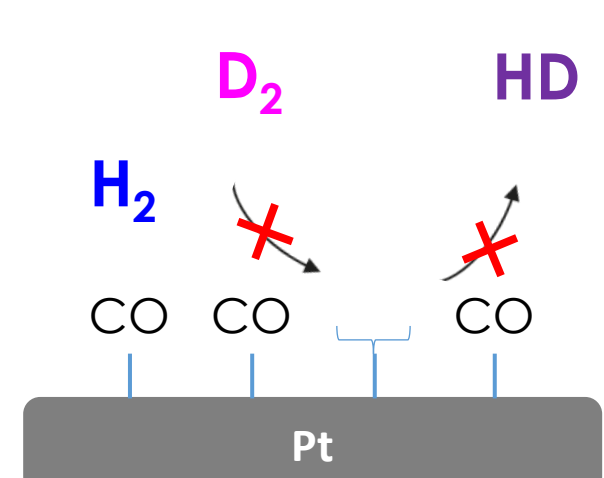
Hindered HD formation



**How many HD turnovers do we have in between
 CO turnovers under light?**

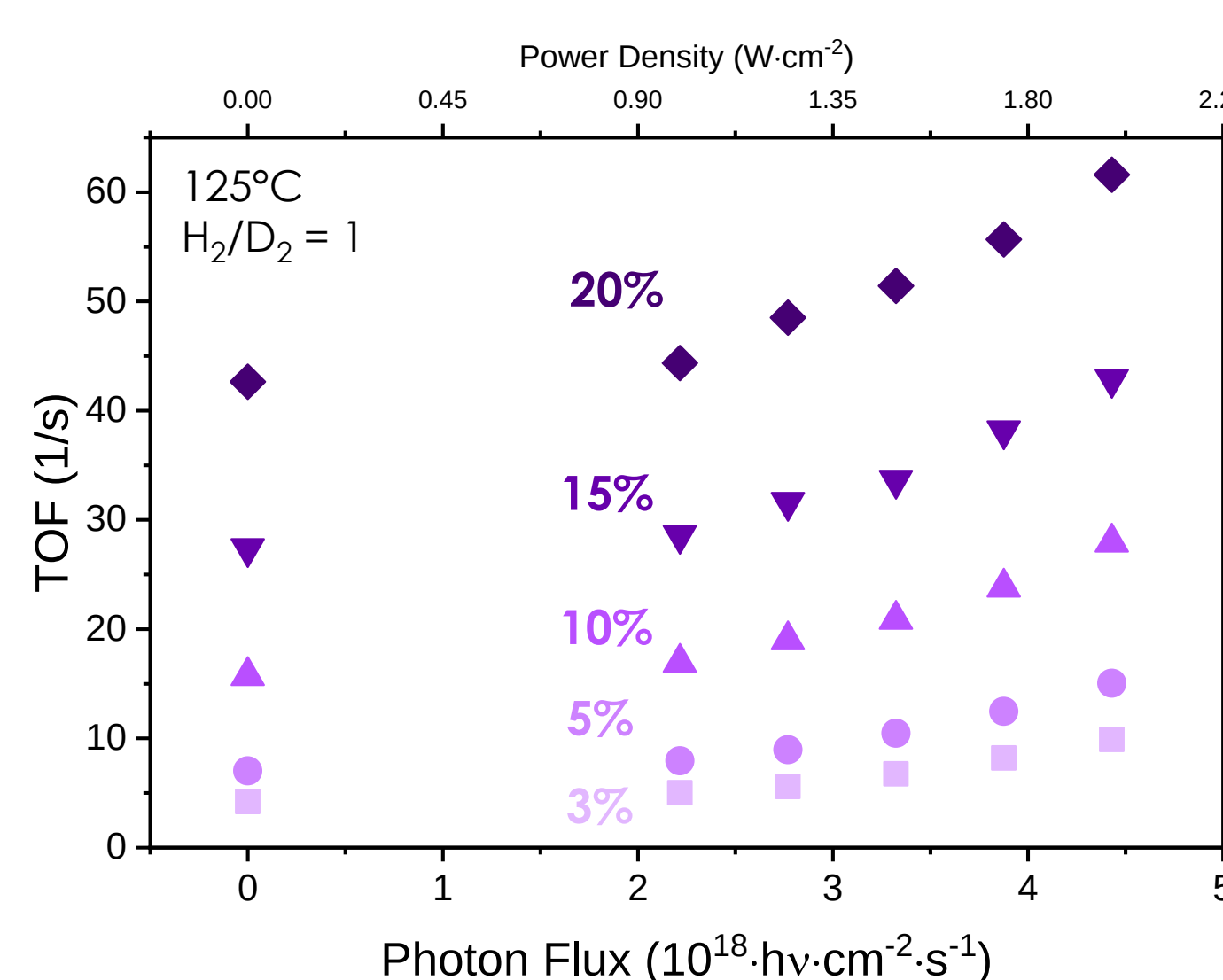
Effect of CO on HD rate

$H_2 + D_2 \rightarrow HD$
with CO co-flow

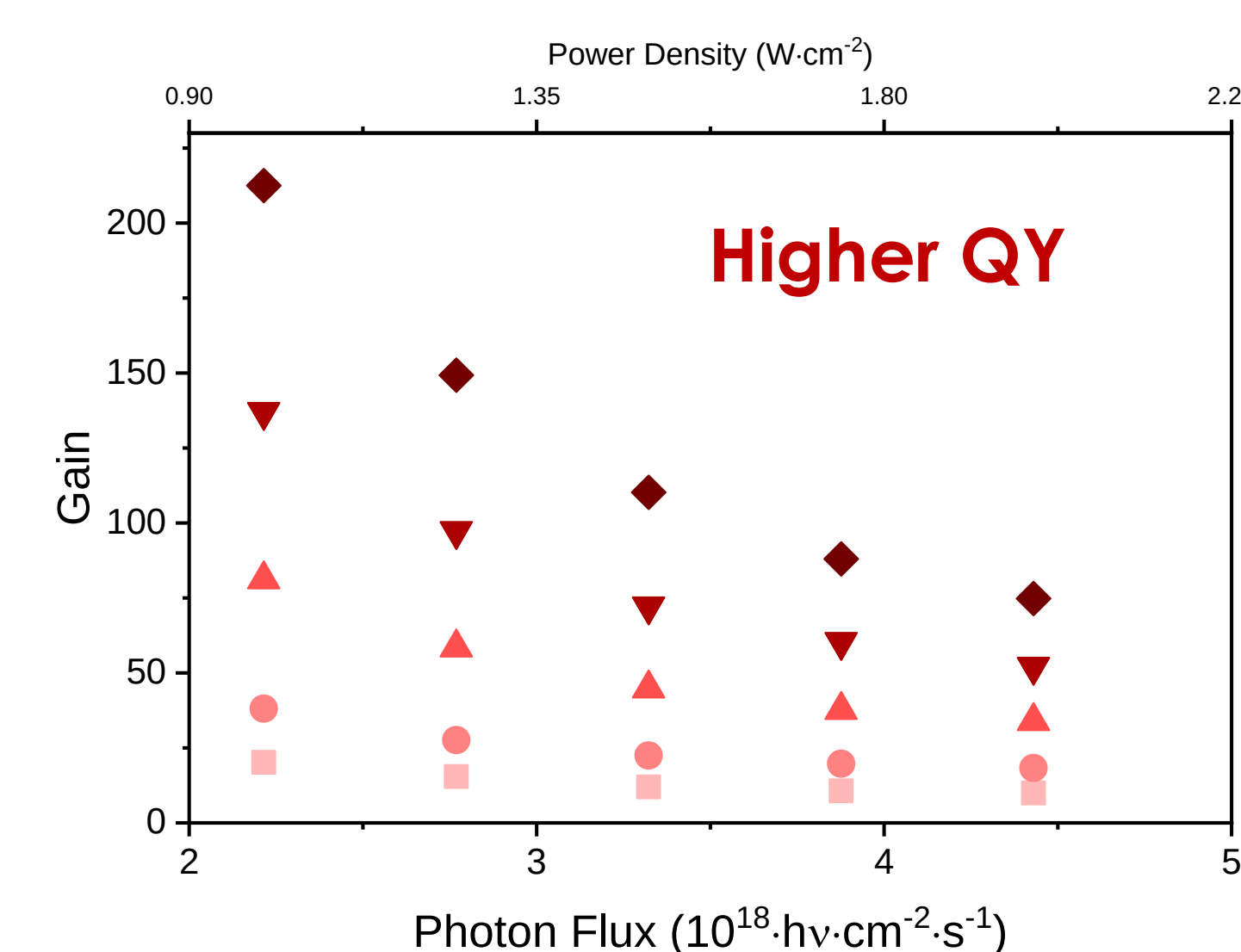


H_2 - D_2 reaction requires pair adjacent sites

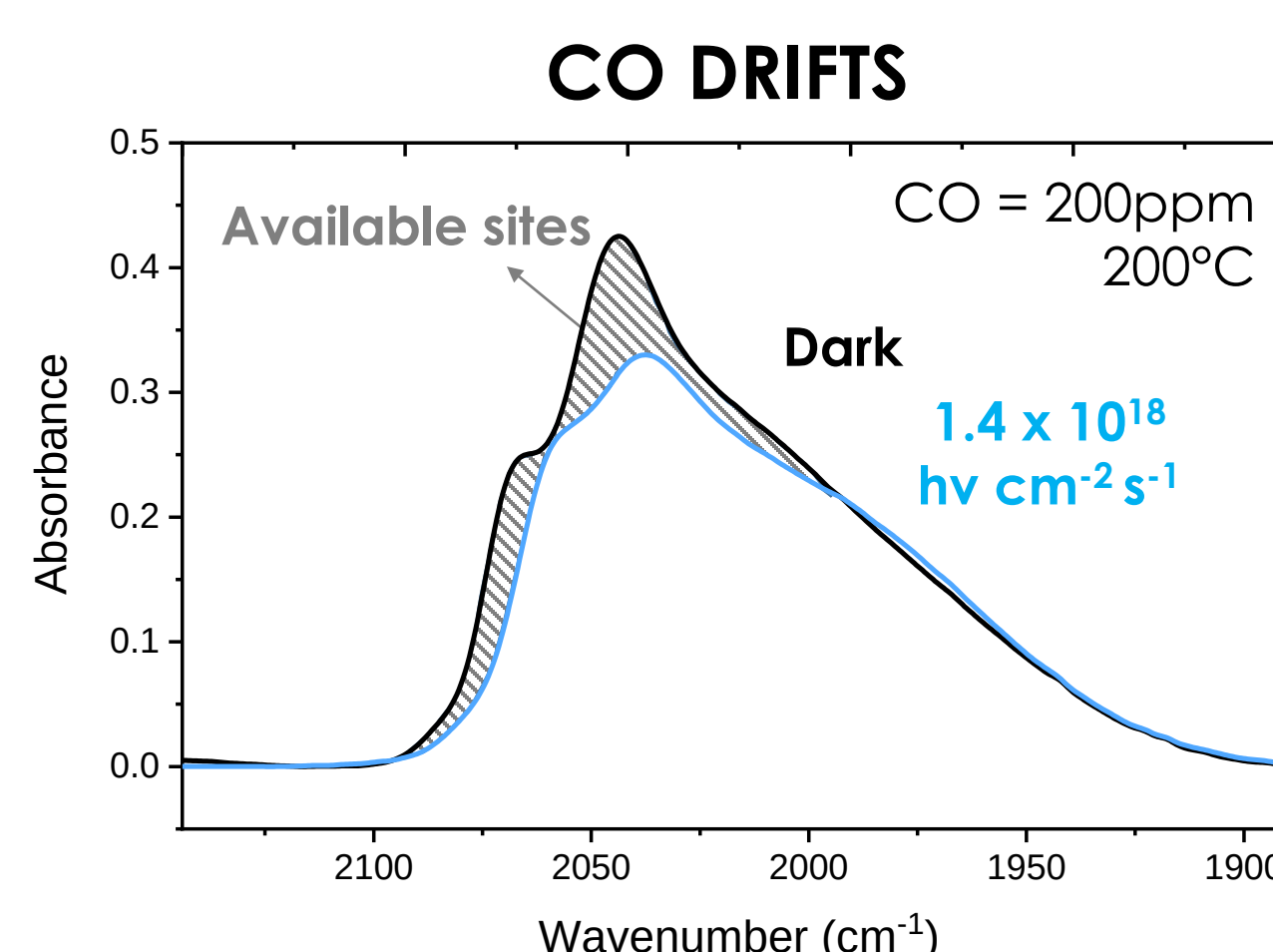
Effect of H_2 and D_2 partial pressure



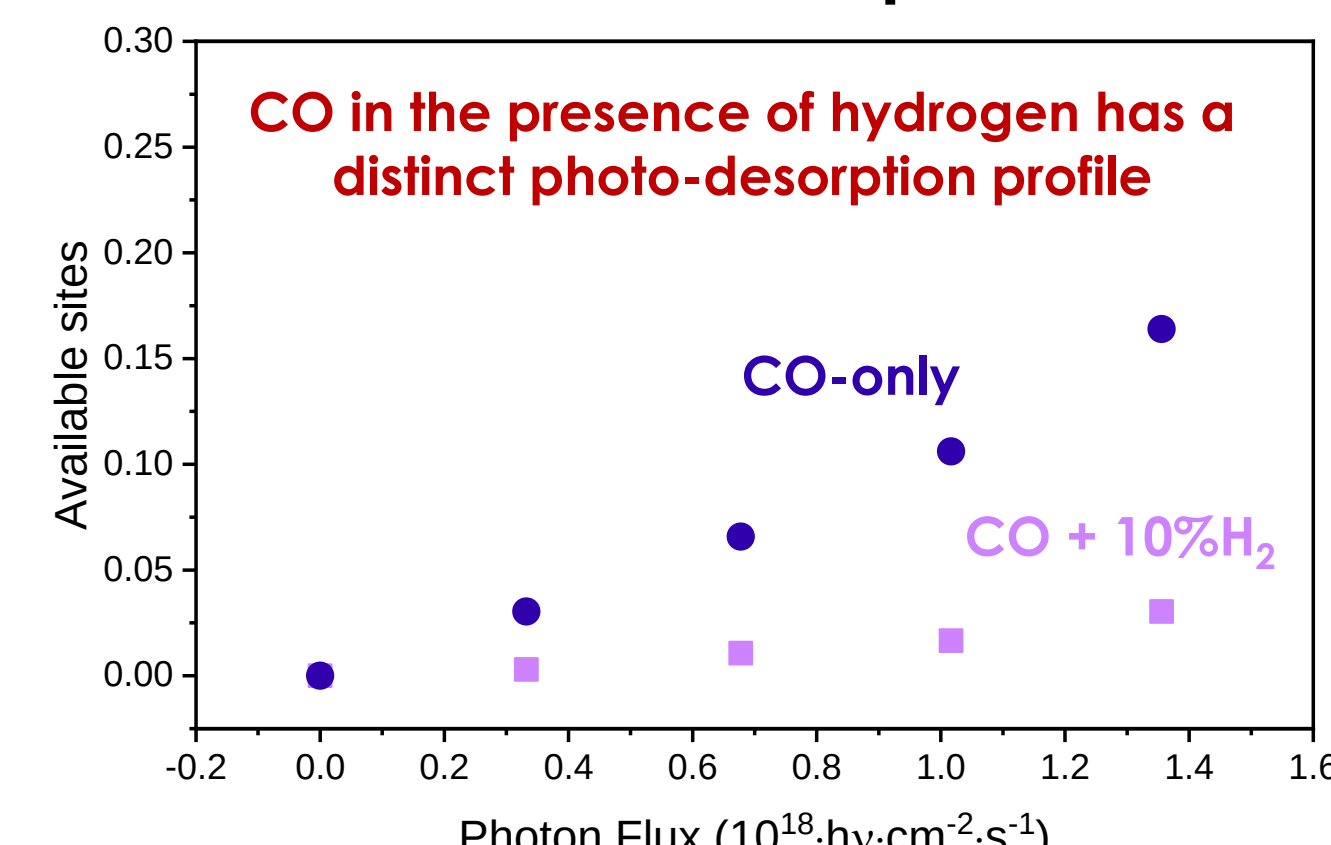
The gain increases with H_2 and D_2 partial pressure



Effect of H_2 on CO coverage upon illumination



Available sites vs photon flux



- New photo-desorption model to describe $CO+H_2$ system
- Estimate the gain with the $CO+H_2$ photodesorption rate

Acknowledgements

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